Lecture 1 part 1: Rough intro to the history of BHs and GWs

October 4, 2023

Warm up "quiz" (feel free to draw something!)

- If the Sun disappeared instantly, would we know instantly?
- Is gravitational free-fall an inertial or non-inertial motion?

Historical overview of BHs

- (1783) John Mitchell calculated that light emitted by a star can be gravitationally trapped if the escape velocity is greater than the speed of light.
  But, this was long before Galilean motion, i.e., $x' = x - \text{vt}$, $t' = \text{t}$,
  was replaced with Lorentzian motion, $x' = \gamma(x - vt)$, $t' = \gamma\left(t - {}^{xv}/_{c^2}\right)$, $\gamma = \frac{1}{\sqrt{1-({}^{v}/_{c})^2}}$
- (1915) Einstein (w/ much help from Marcel Grossmann) published an early form of general relativity (GR), predicting the relativistic correction to the perihelion precession of Mercury, the bending of light by gravitating mass, and gravitational redshift.
- (1916) Karl Schwarzschild (Schw.) published the first nontrivial, analytic solution of Einstein's equations: a non-rotating (spherically symmetric) black hole (BH) characterized by a point mass at the physical singularity and an event horizon (coined by Wolfgang Rindler) located at the Schw. radius.
  - Schwarzschild first wrote his solution in a letter to Einstein while serving in WW1 on the Russian front, which he concluded, "As you can see, the war treated me kindly enough, in spite of the heavy gunfire, to allow me to get away from it all and take this walk in the land of your ideas."
- The Schw. solution was met with much skepticism, being philosophically displeasing and mathematically dubious.
- (1939) Einstein tried to explain that the Schw singularities do not exist in physical reality, but failed to provide convincing arguments regarding stellar gravitational collapse.
- In hindsight, this is ironic considering that the modern theory of gravitational collapse were developed around the same time (i.e., the Tolman-Oppenheimer-Volkoff equation).
- Simultaneously, cosmology was being revolutionized (see eg A. Friedmann), and singularities also appeared in models of the Universe, such as the hot 'big bang' models pioneered by Georges Lemaitre.
- Thus, singularities in GR gradually gained acceptance, but their fundamental nature is still **unknown**.
- (1963) Roy Kerr provided the analytic solution for a rotating (axially symmetric) BH which is a two-parameter family of solutions: the mass and the spin angular momentum of the BH.
- After more than half a century of subsequent work by countless many, BHs are now at the forefront of astronomy and astrophysics. BHs have been observed in nature as the end products of dying stars and as the occupiers of galactic centers.

Historical overview of GWs

- (1893) Oliver Heaviside used an analogy b/w Newton's and Coulomb's laws to suggest that the gravitational force could propagate, but was unsure.

- (1905) Henri Poincare suggested that such 'gravitational waves' (GWs) must propagate at the speed of light to obey Lorentz invariance.

- (1916) Einstein presented GWs in a weak-field linearized version of GR, but it took decades for others to advance the concept (see e.g. Eddington,

- Two big conferences -Berne 1955 and Chapel Hill 1957 - to settle pressing theoretical issues.
  --> E.g., are GWs a form of radiation? ….. Yes: gravitational energy is transported across spacetime and does work on matter, however the GWs have an incredibly weak amplitude

- (1960s) Joseph Weber tried repeatedly to find GWs with his resonant-mass detectors, but his findings were dubious. Nevertheless, his failures helped to motivate others in the hunt for GWs.

- (1974) Hulse and Taylor showed the decrease of the orbital period of the first-ever-observed binary pulsar is explained by GW emission as predicted by GR.

- The theory of GWs in curved background spacetimes, ie with sources of gravity, advanced significantly in parallel with the development of new detection techniques that rely on laser interferometry (e.g. R. Weiss 1975).

- (1980) Kip Thorne generalizes various multipole expansions of the GW waveform for many types of sources.

- During this time, three main approaches developed to help compute reliable waveforms for astrophysical sources. Each approach has its own niche depending on its strengths/weaknesses.

- The development of theoretical methods for modelling and simulating sources of GWs received much interest since the 1960s. This accelerated greatly in anticipation of LIGO.

- The merger process of a newly formed compact binary can be cut in three segments: the pair inspirals for ~Gyr (wide-separation, weak-field, slow-speed limit), then they merge (very strong gravity regime), and then the resultant compact object vibrates (quasi-normal ringdown if BH; extreme matter modes if NS).

  Three frameworks correspond to the three segments of the merger process, i.e., for binary black holes (BBHs):

  - Post-Newtonian approx.: computationally cheap, valid in weak regime; (S. Chandrasekhar 1969, etc.)
  - Numerical Relativity: computationally expensive, valid in all regimes; (Pretorius, Baker, Campanelli, 2005, etc.)
  - Ringdown Spectroscopy: computationally reasonable, valid post-merger (Vishveshwara, Chandrasekhar, Leaver, Berti, etc.)

  The synthesis of these three frameworks enables the complete theoretical construction of full BBH inspiral-merger-ringdown waveforms.

Main references: (feel free to request more)

Earman, 1999

Kennefick, 2007 & 2014

Chen, 2016

Cervantes-Cota, 2016